Next-day smog forecast and advisory for Pakistan

HackInsight 2026 (Data-to-Decision) · Team XYZ (Laksh Kumar, Syed Mustehsan Akhtar Kazmi, Noor Fatima, Amna Iqbal) ·
Dashboard: https://huggingface.co/spaces/lazypanda21/smog-forecast-pakistan · Code: https://github.com/lazy-laksh/hackinsight-smog-forecast

17.7%
lower error than persistence (test month, 80 city-days)

91%
of real values inside the calibrated 80% range

24/7
hourly pipeline, quality gates, public run history

22/22
golden questions passed by the cited Insight Agent

Problem and user

Smog decisions in Punjab and KP (school timings, outdoor activity, traffic and construction limits, hospital readiness) are made too late. A district environment officer needs, **at 20:00**, tomorrow's PM2.5 and a clear recommendation. We forecast tomorrow's mean PM2.5 for Lahore, Islamabad/Rawalpindi, Peshawar and Karachi.

Data and pipeline

- OpenAQ sensors (202 sensors, 905,340 hourly readings since June 2025), NASA FIRMS fires (285,691 detections), Open-Meteo weather and CAMS.
- 2,817 raw files, each with a SHA-256 fingerprint in `data/manifest.csv`, re-verified every run.
- Prefect flow every hour: fetch → verify → clean → Data Quality Agent (gate 1 schema, gate 2 freshness and completeness) → gold → forecast → publish. A failed gate stops the run; every quality decision is logged (1,552 sensor-hours flagged: stuck, spikes, zeros, impossible values).

Model

Gradient-boosted trees on what is known at 20:00 PKT (today so far, recent days, weather forecast, fires, CAMS). Settings chosen by month-by-month rolling-origin validation; target days from 2026-10-01 were locked as the test set. Declared baseline: persistence (today so far).

City (test month)	Our MAE	Persistence	Better	80% range
Islamabad/Rawalpindi	6.37	10.53	+39.5%	90%
Karachi	4.87	4.14	-17.6%	90%
Lahore	12.74	13.95	+8.7%	90%
Peshawar	11.8	14.82	+20.4%	95%
All	**8.94**	**10.86**	**+17.7%**	**91%**

Declared success (≥10% better overall and in Lahore): overall met, Lahore NOT met. Strict model without any "tomorrow" inputs: 7.5% better. Ranges widened by 5.66 µg/m³, chosen on validation months only (coverage 59% → 80%).

From forecast to decision

The calibrated range gives a **chance of an unhealthy day** (≥ 55.5 µg/m³). The dashboard recommends an advisory when that chance ≥ 1/(1+R), where R (slider, default 3) is how much worse a missed smog day is than a false alarm. Replayed on days the model never trained on:

- validation months: caught **614/632** unhealthy days with 183 false alarms (persistence rule: 558/632, 69 false alarms)
- test month: caught **39/40** unhealthy days with 17 false alarms (persistence rule: 29/40, 7 false alarms)

Probability skill vs the base rate (Brier, validation): 0.72. Each forecast also shows its main reasons (smog already in the air, weather, fires, season) and an Urdu advisory to forward.

Insight Agent (RAG)

Retrieves from cited official sources (WHO 2021, US EPA AQI 2024, Pakistan NEQS 2010, Punjab EPD and P&D, school instructions 2026) and from data facts built by code from gold rows. An LLM (Groq gpt-oss, Gemini fallback) answers; our code then checks every citation id and every number against the sources, retries once, and otherwise answers *"I don't know based on my sources."* Personal details are removed before the model sees a question; health questions always get a "call Rescue 1122" note. English and Urdu. Golden set: 22/22 passed.

Dashboard and operations

Free static Hugging Face Space reading the hourly published dataset: last successful run, data freshness, quality gates, forecasts, live "were we right?" report card, drift monitor (inputs vs the same weeks last year, live error vs test error, every run), back-test, advisory replay, model card. Checked against Punjab EPA reference stations on 4 hours: our Lahore value was on average 10 µg/m³ lower than the EPA station mean (ratio 0.89).

Business model and impact

Free public dashboard and Urdu bulletin; paid decision service (alerts at the user's own threshold, API, more cities) for district administrations, school networks, hospitals and employers of outdoor workers. Running cost: one small machine; a commercial Open-Meteo plan is needed before charging.

Limitations

Low-cost sensors (no humidity correction); one value per city; archived weather used for "tomorrow" in training; Karachi does not beat persistence on the test month; in last winter's smog season (validation) the model was worse than persistence in Islamabad/Rawalpindi, Lahore, Peshawar (worst: Lahore -52%), because the data starts in June 2025 and trees cannot predict above levels they have not seen; the final model has now seen last winter; the default rule trades more false alarms for fewer misses. Information, not medical advice.